GRAPHICAL ABSTRACT: Governing Human–AI Collaborative Project Workflows in Multi-Platform Digital Delivery

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1. UNGOVERNED MULTI-AGENT RISKS

Handoff Drift & Role Ambiguity

- Asynchronous agents (GPT, Claude, DeepSeek) overwrite shared state
- Out-of-window autonomous writes
- Missing ownership & acceptance gates

Non-Idempotent Retry Faults

- Hybrid REST API & Browser DOM timeouts
- 92.32% latency in post-submit wait
- Naive retries trigger duplicate posts
- CAPTCHA / expired cookies crash flow

Epistemic & Metric Confounding

- Exposure-time bias across cohorts
- Legacy (3,477 views) vs AI series (95.89 views, $p = 0.0012$)
- Unverified causal attribution claims

2. FOUR-LAYER STAGE-GATE GOVERNANCE ARCHITECTURE

Layer 1: Sole-Operator Authority & Deterministic Scheduling Gate (Alg. 1: planRun)

- Role partitioning: Human PM (Sole Operator) + GPT/Codex (QA) + Claude (UI) + DeepSeek (Monitor)
- Pure predicate $G_{run} = I_{enabled} \wedge I_{window} \wedge I_{cooldown} \wedge I_{checklist}$ (Thm A.1: 0 unauthorized writes)

Layer 2: Cryptographic Plan-Hash Locking & Stream SHA-256 Gate (Theorem A.3)

- 64-KB chunked SHA-256 digest $H(f)$ + deterministic JSON plan hash $H_{plan} = SHA256(Canon(M, T, H(f)))$
- Execution blocked unless token matches (--confirm CURRENT_PLAN_HASH); locks 30 artifacts (2.21 GB)

Layer 3: Idempotent Receipt State Machine & UNKNOWN Latching (Alg. 2 & Thm A.2)

- States: INIT → DRY_RUN → PREFLIGHT → SUBMITTING → {PUBLISHED, UNKNOWN, NEEDS_HUMAN, FAILED}
- Post-submit read-back timeout latches receipt in UNKNOWN & suppresses auto-retries (At-Most-Once External Publication under $K \geq 1$ recovery rounds; prevented duplicate post on Douyin)

Layer 4: Read-Back Verification, HITL Handoff & Epistemic Audit Gate (Alg. 3)

- Independent URL/ID read-back verification + structured NEEDS_HUMAN handoff for CAPTCHA / auth expiry
- Mann–Whitney U cohort stratification ($N = 55$) + 4 mandatory epistemic hygiene rules

3. EMPIRICAL VALIDATION & EVIDENCE

Longitudinal Multi-Agent Governance (N = 69)

- 68 non-write checkpoints blocked (100% gate accuracy)
- 1 authorized write; 0 duplicate replays; 0 new risks
- Heartbeat delay: 24.992 s | Run time: 31.175 s

5-Platform Hybrid Release Telemetry

- Bilibili, YouTube, X, Douyin, Instagram verified
- Instagram E2E: 423.236 s (380.551 s / 92.32% in post-submit wait window → primary retry hazard)

Software & Media Verification Scale

- 123 Python source modules (27,095 LOC)
- 126 test modules (37,155 LOC; 2,737 unit tests)
- 30 SHA-256 locked video releases (2.21 GB)

Reproducible Open-Science Compendium

- 21-Page Main Manuscript + 150-Page Online Supplementary Material (Appendices A–H)